African Savannah with AI

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Introduction

This assignment compares a handwritten Savannah simulation with an AI-assisted extension in GAMA. The focus is on preserving specified rules while adding zebra grazing and lion energy dynamics.

Methods – including specification

The model uses a continuous 100 × 100 m world with 20 lions and 20 zebras at initialization. Both species wander. Lions kill nearby zebras within 2 m. A 50 × 50 grass grid begins with 0–10 kg per cell and grows by 0.1 kg per step. In the extension, zebras graze up to 1 kg per step; lions start with 50 energy, lose 0.05 per wandering step, gain 10 per zebra killed, and starve at zero. The model includes map, population and grass-biomass displays.

Results and screenshot interpretation

The screenshot shows a later simulation state with **20 lions and 4 zebras**. The population plot indicates that zebras decline from 20 to 4 over roughly 1,600 steps, while lions remain at 20. The grass-biomass chart rises almost linearly to approximately **160 kg per cell**. These observations are consistent with continued grass growth despite grazing, and show that predation reduces zebra abundance in this run.

Discussion

The screenshot provides visual evidence that the model runs and records population and vegetation dynamics, but does not by itself verify every rule or demonstrate ecological realism. In particular, the unchanged lion count suggests that lion starvation did not occur during the displayed run; the increasing grass biomass suggests that growth outweighs grazing and is not capped at 10 kg. Outcomes depend on parameters and random encounters, so repeated runs would be needed to assess robustness. The specification, code comparison and AI log document the implementation and its limitations.

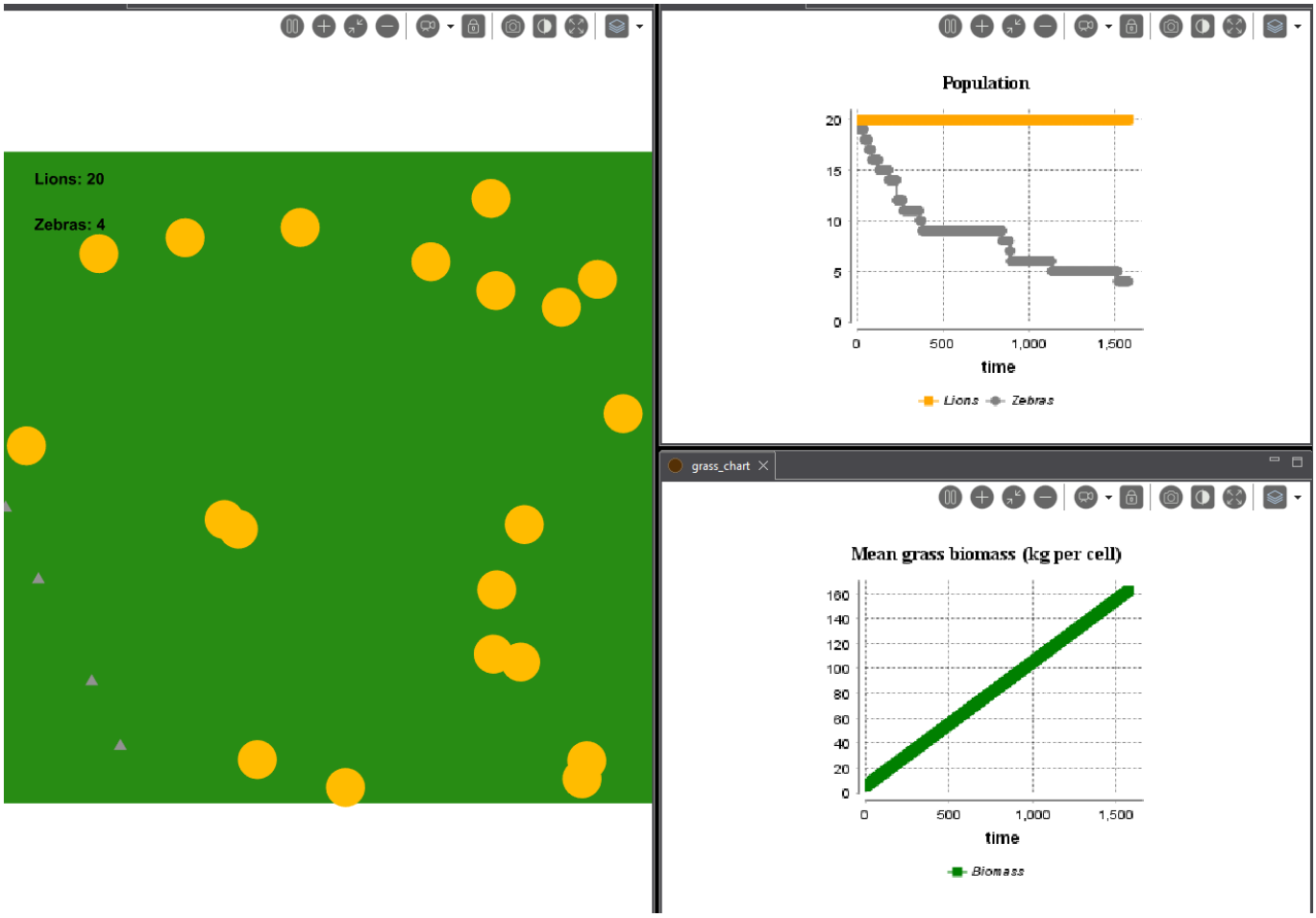


Figure 1. GAMA Savannah simulation (left), lion and zebra populations (top right), and mean grass biomass (bottom right), at approximately 1,600 simulation steps.